Chemistry for Materials/Polymer Chemistry

Question Set 3

Comments: some questions in the question set are not like exam or test questions. They are used for the purpose of reflecting on some topic to ensure students understand the main concepts. These will be called here "Reflecting questions" (RQ). The questions more similar to exam/test questions will be labelled ETQ.

In class questions:

1) Have a look at the Lecture Notes Set 2 and ask questions of what you do not understand.

2) Imagine two monomers A and B. A has a structure that enables it to polymerise through chain-growth polymerisation, while B has a structure that enables it to polymerise through step-growth polymerisation. (The mechanism or structure is really not important for this question). Scenario 1: Imagine you start polymerising A, and half-way through the polymerisation, you add C, also another monomer, which can polymerise through chain-growth polymerisation, and, that can polymerise together well with A (this is called "copolymerise"). Separately, imagine Scenario 2: you start polymerising B, and half-way through the polymerisation, you add D, a monomer, which can polymerise through step-growth polymerisation, and, that can copolymerise well with B. At the end of both polymerisation processes, what type of polymer samples will you have in each Scenario? (You may want to think of the polymers as A-A-A-A….-or A-A-C-C, etc, or combinations thereof). If you understand this: you will understand the differences between chain-growth and step-growth polymerisations.

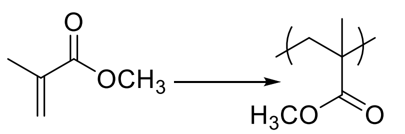
   RQ: A + C: AAAAAAAAAAAAAA…. + AACACCAAACCACACCAAA… (random?) (if A is depleted before C, THEN, we can see the formation of CCCCCCCCCCCCCCC…..)

   B+D: BBBDDBBBDDDBD (random) (+rings BD) (maybe rings of B, but not rings of D alone).

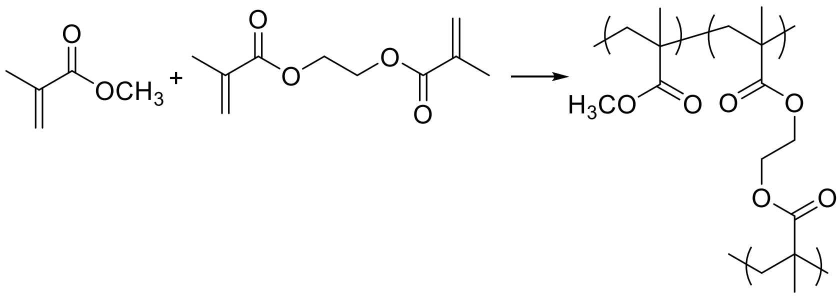
   Difference: chain growth: you have polymer formed early (so, AAAAAAAAAAAAA… can form before you add C and maybe CCCCCCCCCCCCCCCC can form after there is no more A). but in step growth we will never see BBBBBBBBBBBBBBBBBBBBB…. (because there won't be polymer of B before you add D, only oligomers of B that will react with D to form a polymer).

3) Classify the following process according to "step-growth" or "chain-growth", and also the 4 types of polymerisation reactions defined by IUPAC. Also determine whether they are hompolymers or copolymers.

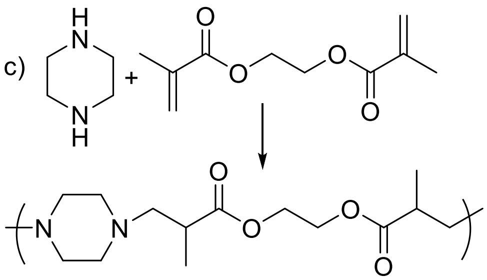
a)



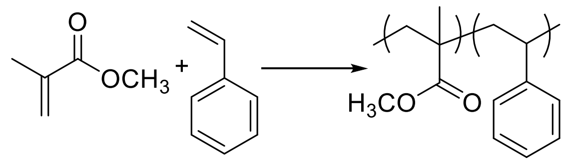
b)



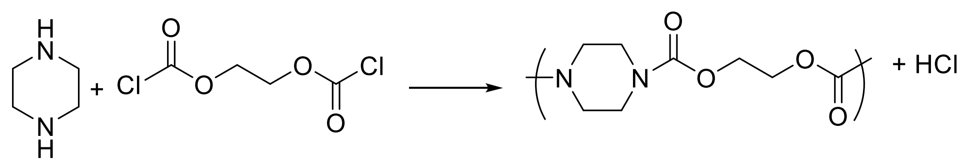
c)



d)



e)



ETQ: a) Chain growth, chain polymerisation. Reason: only one functional group per monomer. Homopolymer.

b) Chain growth, chain polymerisation. Reason: only one functional group per monomer gets incorporated into a single chain (there is a monomer with 2 groups, but they form part of DIFFERENT chains). Copolymer (two different monomers).

c) Step-growth, polyaddition. Reason: there are two functional groups per monomer and they all form part of the same chain. There are two different monomers, but we don't normally call this a copolymer. This is because BOTH monomers are needed to form a polymer. If you use only one monomer you cannot get a polymer.

d) Chain growth, chain polymerisation. Reason: only one functional group per monomer. Copolymer (two different monomers).

e) Step-growth. Polycondensation. Reason: there are two functional groups per monomer and they all form part of the same chain. There are two different monomers, but we don't normally call this a copolymer. This is because BOTH monomers are needed to form a polymer. If you use only one monomer you cannot get a polymer.

4) Consider a monomers A with two functional groups B and C that polymerise by <u>Polyaddition</u>. Knowing that MW (A) = 180 g/mol determine the maximum M_n that can be achieved in the following cases (use suitable equations to produce your answer):

   a) Conversion: 95%

      ETQ: A full answer for full marks in an exam will explain the following: because this is polyaddition, as it is step-growth, you can use the Carothers equations to determine degree of polymerisation (DP). To calculate the Mn from DP you need the MW of the repeating unit, but you do not know the structure of the polymer. However, polyaddition means ALL the atoms in the monomer will be part of the repeating unit, therefore, in this case the MW of the monomer = MW repeating unit (180 g/mol).

Applying Carothers equation (show equation and explain the variables):

$$\overline{DP} = \frac{1}{1-p}$$

Conversion 95%: p=0.95, therefore (show calculation): DP=20, Mn=20*180=3600

b) The system has an impurity X that contributes more functional group B, but not C. In other words, X is not a monomer. Molar % : A = 97%, X = 3%, conversion: 100%

ETQ: for full marks you need to identify that the impurity produces a stoichiometric imbalance. Calculate the stoichiometric imbalance using the equation (shown for the lecture notes, of course, you have to calculate for B and C, not A and B): as r<1, $r=N_C/N_B$ (r=97/100, r=0.97)

number)

$$r = \frac{N_A^o}{N_B^o}$$

Always defined

$$r \leq 1$$

$$N_B^o \geq N_A^o$$

For full marks you need to explain the need to use the general Carothers equation, show the equation and explain the variables.

$$\boxed{\overline{DP} = \frac{1+r}{1+r-2rp}}$$

Now show the equation and the calculation with r=0.97 and p=1

DP=65.7 (Mn=65.7*180=11820).

c) The system has an impurity X that contributes more functional group B, but not C. In other words, X is not a monomer. Molar % : A = 97%, X = 3%, conversion: 95%

ETQ: for full marks you need to identify that the impurity produces a stoichiometric imbalance. Calculate the stoichiometric imbalance using the equation (shown for the lecture notes, of course, you have to calculate for B and C, not A and B): as r<1, $r=N_C/N_B$ (r=97/100, r=0.97) number)

$$r = \frac{N_A^o}{N_B^o}$$

Always defined

$$r \le 1$$

$$N_B^o \ge N_A^o$$

For full marks you need to explain the need to use the general Carothers equation, show the equation and explain the variables.

$$\boxed{\overline{DP} = \frac{1+r}{1+r-2rp}}$$

Now show the equation and the calculation with r=0.97 and p=0.95

DP=15.5 (Mn=15.5*180=2792).

d) What would your answer be, and why, if the monomers polymerise by Polycondensation?
RQ: Polycondensation is also step-growth, so Carothers equations are valid. BUT, you can only calculate DP and not Mn because the Mw repeating unit is unknown.

e) What would your answer be, and why, if the monomers polymerise by chain-polymerisation?
RQ: Chain polymerisation is a chain-growth polymerisation, so Carothers equations are NOT valid. This means, you don't have data to calculate anything relating the MW.